

DPP-2

- Q.1 If the three points with position vectors $(1, a, b)$; $(a, 2, b)$ and $(a, b, 3)$ are collinear in space, then the value of $a + b$ is
(A) 3 (B) 4 (C) 5 (D) none
- Q.2 Consider the following 3 lines in space
 $L_1 : \vec{r} = 3\hat{i} - \hat{j} + 2\hat{k} + \lambda(2\hat{i} + 4\hat{j} - \hat{k})$
 $L_2 : \vec{r} = \hat{i} + \hat{j} - 3\hat{k} + \mu(4\hat{i} + 2\hat{j} + 4\hat{k})$
 $L_3 : \vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k} + t(2\hat{i} + \hat{j} + 2\hat{k})$
Then which one of the following pair(s) are in the same plane.
(A) only L_1L_2 (B) only L_2L_3 (C) only L_3L_1 (D) L_1L_2 and L_2L_3
- Q.3 The acute angle between the medians drawn from the acute angles of an isosceles right angled triangle is:
(A) $\cos^{-1}(2/3)$ (B) $\cos^{-1}(3/4)$ (C) $\cos^{-1}(4/5)$ (D) none
- Q.4 If $\vec{e}_1$ & $\vec{e}_2$ are two unit vectors and θ is the angle between them, then $\cos(\theta/2)$ is
(A) $\frac{1}{2}|\vec{e}_1 + \vec{e}_2|$ (B) $\frac{1}{2}|\vec{e}_1 - \vec{e}_2|$ (C) $\frac{\vec{e}_1 \cdot \vec{e}_2}{2}$ (D) $\frac{|\vec{e}_1 \times \vec{e}_2|}{2|\vec{e}_1||\vec{e}_2|}$
- Q.5 The vectors $3\hat{i} - 2\hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} + 5\hat{k}$ & $2\hat{i} + \hat{j} - 4\hat{k}$ form the sides of a triangle. Then triangle is
(A) an acute angled triangle (B) an obtuse angled triangle
(C) an equilateral triangle (D) a right angled triangle
- Q.6 If the vectors $3\vec{p} + \vec{q}$; $5\vec{p} - 3\vec{q}$ and $2\vec{p} + \vec{q}$; $4\vec{p} - 2\vec{q}$ are pairs of mutually perpendicular vectors then $\sin(\widehat{\vec{p} \vec{q}})$ is
(A) $\sqrt{55}/4$ (B) $\sqrt{55}/8$ (C) $3/16$ (D) $\sqrt{247}/16$
- Q.7 Consider the points A, B and C with position vectors $(-2\hat{i} + 3\hat{j} + 5\hat{k})$, $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $7\hat{i} - \hat{k}$ respectively.
Statement-1: The vector sum, $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$
because
Statement-2: A, B and C form the vertices of a triangle.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.
- Q.8 The set of values of c for which the angle between the vectors $c\hat{i} - 6\hat{j} + 3\hat{k}$ & $x\hat{i} - 2\hat{j} + 2cx\hat{k}$ is acute for every $x \in \mathbb{R}$ is
(A) $(0, 4/3)$ (B) $[0, 4/3]$ (C) $(11/9, 4/3)$ (D) $[0, 4/3]$
- Q.9 Let $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u} \cdot \hat{n} = 0$ and $\vec{v} \cdot \hat{n} = 0$, then $|\vec{w} \cdot \hat{n}|$ is equal to
(A) 1 (B) 2 (C) 3 (D) 0
- Q.10 If the vector $6\hat{i} - 3\hat{j} - 6\hat{k}$ is decomposed into vectors parallel and perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ then the vectors are :
(A) $-(\hat{i} + \hat{j} + \hat{k})$ & $7\hat{i} - 2\hat{j} - 5\hat{k}$ (B) $-2(\hat{i} + \hat{j} + \hat{k})$ & $8\hat{i} - \hat{j} - 4\hat{k}$
(C) $+2(\hat{i} + \hat{j} + \hat{k})$ & $4\hat{i} - 5\hat{j} - 8\hat{k}$ (D) none